Reuben Seyram Komla Agbozo

User Behavior Architect · Behavior analysis, human factors and human–AI interaction

USER BEHAVIOR RESEARCH · BEHAVIORAL DATA ANALYSIS · AI-ENABLED PROTOTYPING

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Hangzhou, China · open to remote work and relocation

Profile

I am a user behavior architect and human factors researcher, focused on how people think, act and adapt within systems. My doctoral research measures how driver workload changes when driving tasks shift to the system: I built the simulator lab, proposed the measurement methodology and ran the study with 51 drivers. I notice what isn't working for people, define the real problem, and build practical solutions: products, prototypes, learning programs and the way an organisation runs. Earlier work includes founding and leading a team from scratch, designing AI curricula and training teachers to use AI, and developing a company's organisational structure. I work across engineering, design and education, and across cultures in English and Mandarin.

> I am looking for user behavior research roles: analysing how people use products, platforms and AI systems, finding where a design works against the user, and proposing changes that serve them. Open to remote work and relocation. References available on request.

Research at a glance

- Built the research group's first driving-simulation lab, with a driver assistance system offering steering, pedal and combined support.
- Proposed the Driver Workload Assessment Methodology (DWAM): objective measurement (visual and physical) and subjective measurement, using Tobii eye tracking, Xsens motion capture and NASA-TLX.
- Ran a within-subject study with 51 drivers across four conditions (manual driving and three assist modes), counterbalanced with a Williams Latin square.
- Guided two master's students' simulator projects to graduation.
- Co-author (second of seven) of Unsupervised identification of driving styles from naturalistic driving data through a context-normalized framework, Sensors (2026).

What I bring

Problem framing
Turns vague, messy problems into clear questions a team can answer and act on.

Human behavior and systems
Designs systems that adapt to how people actually behave, instead of asking people to adapt to the system.

Building and testing
Builds working prototypes quickly, so ideas are tested before time and money are committed.

Analytical judgment
Combines hard data with what people report to reach conclusions that hold up under scrutiny.

Learning new fields fast
Becomes productive quickly in unfamiliar fields and carries ideas from one domain into another.

Critical review
Gives clear, fair assessments that help others improve their work.

Auditing and quality systems
Checks whether processes deliver what they promise and finds where they fall short.

AI learning and communication
Helps people and teams adopt AI, and explains technical ideas in terms anyone can use.

Leadership and organisation
Builds teams and structures that make work clear, measurable and efficient.

Methods and tools

Study design and analysis	Within-subject experiments; counterbalancing; repeated-measures and mixed-effects analysis; correlation, multiple-testing correction and sensitivity analysis.
Human-factors measurement	Tobii eye tracking; Xsens motion capture; NASA-TLX (a standard rating scale for perceived workload); synchronized multimodal data processing and interpretation.
Simulation and computation	CARLA; CarSim; model predictive control (MPC); Logitech steering and pedal interfaces; Python analysis pipelines; vehicle telemetry; machine vision with Mask R-CNN; Neo4j knowledge graphs.
AI and educational design	AI-assisted prototyping; web-app and agent workflows; curriculum development; assessment design; teacher training.
Auditing and compliance	ISO 9001, 14001 and 45001 gap analysis and auditing; management system design.

Research and professional experience

SEP 2021 TO PRESENT *Jan 2024 to present*

Doctoral Researcher, Human–Intelligent Vehicle Interaction

Zhejiang University, Hangzhou

WHAT I DID

- Built the group's first driving-simulation lab, including a driver assistance system with steering, pedal and combined support.
- Proposed the Driver Workload Assessment Methodology (DWAM): objective measurement (visual plus physical) and subjective measurement, for each driving mode, showing whether, where and in which direction workload changes as driving tasks shift to the system.
- Guided two master's students in designing and building their experiments and analysing their data to answer their research questions.

OUTCOMES

- The lab now supports the group's ongoing and future studies.
- Proposed to system designers that driver workload be added to the SAE levels of driving automation (the standard scale from no automation to full automation), which define only what the driver must do, together with a methodology to quantify it. The study had 51 drivers, each completing four conditions in the simulator: manual driving and steering, pedal and combined support.
- Both master's students graduated successfully and are now employed.

SEP 2021 TO DEC 2023 *Including Research Assistant with Siemens, Sep 2022 to Aug 2023*

Doctoral Researcher, Smart Sustainable Manufacturing

Zhejiang University, Hangzhou · Siemens, Shanghai

WHAT I DID

- Proposed the concept of cognitive intelligence-enabled manufacturing.
- Investigated causality-based quality problem solving for discrete manufacturing systems.
- With Siemens: built a data-driven Neo4j knowledge graph of low-carbon manufacturing processes to support automated life cycle assessment.
- With Siemens: studied how big data can support greener, smarter production across the production value chain.

OUTCOMES

- First-author paper, “Towards Cognitive Intelligence-Enabled Manufacturing,” APMS 2022.
- Co-authored “Two-stage approach to causality analysis-based quality problem solving for discrete manufacturing systems”: Best Student Conference Paper, NSFC-RGC 2023 Conference, Hong Kong. Extended version published in the Journal of Engineering Design (2025).

APR 2024 TO PRESENT
PART-TIME ADVISORY

Program Research Lead and Advisor

Emerald Hill Design Education Institute, Hangzhou

WHAT I DID

- Designed the institute's curriculum on AI, web apps and AI agents.
- Helped teachers bring AI into their teaching.
- Advised leadership on research and management.

OUTCOMES

- Institute revenue increased by 18% during my role.
- About 30 students and working professionals trained.

NOV 2024
FREELANCE

Management Consultant

Komodo Mining Services, Ghana

WHAT I DID

- Restructured the company's service offerings into a clear, marketable set of services.
- Designed its people management system, including how work is reported and quantified.

OUTCOMES

- The new service structure helped the company secure overseas partnerships.
- Consistent reporting and measurable work output across the team.

JUN TO AUG 2025
FREELANCE

AI and Robotics Curriculum Consultant

Belio Technology (a subsidiary of WaShalimba Group), Namibia

WHAT I DID

- Designed progressive AI and robotics curricula for four learner groups: ages 9 to 11, 12 to 15 and 16 to 18, and adult and professional learners.

OUTCOMES

- The client used the curriculum outline in its application for funding to set up an AI and robotics learning centre in Windhoek, Namibia.

MAY TO AUG 2026
FREELANCE *Credential: Internal Auditor, ISO 9001, 14001 and 45001, BSI Training Academy, Singapore (2020)*

QHSE Advisory Consultant

IgnisOak Advisory, Ghana

WHAT I DID

- Audited clients' management systems for gaps before industrial safety and thermal risk engineering solutions were deployed.
- Aligned engineering controls with ISO 45001, audited environmental and emissions systems under ISO 14001, standardised quality documentation under ISO 9001, and designed and audited the firm's own integrated management system.

OUTCOMES

- Helped clients restructure their organisational management systems.
- The firm now has its own integrated management system, ready for enterprise-level compliance.

AUG 2020 TO AUG 2021

Research Assistant

Lanzhou University of Technology, Lanzhou

WHAT I DID

- Continued my master's research on steel microscopy images, exploring alternative approaches to the Mask R-CNN method I developed to locate, segment and measure carbide particles. Assisted supervisors with research tasks, including conference organisation.

OUTCOMES

- The Mask R-CNN method located and segmented over 90% of target particles.
- First-author paper, Journal of the Korean Society for Precision Engineering (2020).

MAY TO JUL 2015
CONTRACT

Product Research Analyst

Akosombo Paper Mill Ltd (APML), Akosombo

WHAT I DID

- Co-led a market intelligence study on the tissue paper industry across Ghana and West Africa, to guide the company's growth strategy after its rebrand.
- Evaluated regional competitors, product lines and buyer behaviour, and developed recommendations that redefined the company's brand structure and positioning.

OUTCOMES

- Produced investor-ready insights that secured external funding and strategic partnerships.
- Laid the groundwork for an industrial paper hub in Akosombo that later attracted over €5M in international expansion from The Good Roll.

Leadership and teaching

Mar 2018 to Aug 2021

Founding Editor in Chief

School of International Education, Lanzhou University of Technology

WHAT I DID

- Founded and led the school's first editorial and media team.
- Created a website, YouTube channel, Facebook and Instagram pages, promotional materials and an overseas partnership program.

OUTCOMES

- Team of about 10 built from scratch.
- Four online channels launched to support international student admissions.

Aug 2015 to Aug 2016

Teaching Assistant

University of Ghana, Accra

WHAT I DID

- Taught engineering students as a lecturer's assistant in algebra, calculus 1 and 2, environmental engineering, and thermodynamics 1 and 2.

OUTCOMES

- Six courses.
- Classes of up to about 200 students.

2009 to 2011

Science and Mathematics Teacher

Stage International School, Ashaiman, Ghana

WHAT I DID

- Taught science and mathematics to primary and middle school students.

OUTCOMES

- Every student passed the Basic Education Certificate Examination (BECE) and gained admission to senior high school.

Education

Sep 2021 to Mar 2027 (expected)

PhD, Industrial and Systems Engineering (Mechanical Engineering)

Zhejiang University

Driver workload assessment and human well-being in smart vehicles.

Sep 2017 to Jun 2020

MEng, Mechanical Manufacture and Automation

Lanzhou University of Technology

Digital image processing for quantitative metallography. Best Graduating Student and Valedictorian.

Sep 2016 to Jun 2017

Chinese Language and Culture

Lanzhou University of Technology

HSK levels 4 and 5 (the standard Mandarin proficiency test).

Sep 2011 to Jun 2015

BSc Engineering, Process Engineering

University of Ghana

Process plant design.

Publications and writing

Peer-reviewed publications, newest first, each linked to its DOI: Unsupervised identification of driving styles from naturalistic driving data through a context-normalized framework, Sensors (2026); Low-frequency-data-driven energy consumption prediction for battery electric vehicles, Energy (2025); Energy-efficient cutting path planning for wire electrical discharge machining of Cr12 steel based on energy benchmarking, Machining Science and Technology (2025); Two-stage approach to causality analysis-based quality problem solving, Journal of Engineering Design (2025); Knowledge graph-based mapping and recommendation to automate life cycle assessment, Advanced Engineering Informatics (2024); Enhancing big data for greentelligence across the production value chain, Green Manufacturing Open (2023); Optimal Load Distribution of CHP Based on Combined Deep Learning and Genetic Algorithm, Energies (2022); Towards Cognitive Intelligence-Enabled Manufacturing, APMS (2022); Quantitative Metallographic Analysis of GCr15 Microstructure Using Mask R-CNN, Journal of the Korean Society for Precision Engineering (2020). Full record: ORCID 0000-0001-8147-7609. The 2023 Hong Kong conference paper has no DOI.

Author of two essays on designing systems around people: "Machines in Service of Minds: Building AI Around People, Not the Reverse" (12 November 2026) and "The Tyranny of the Default: Toward a User Behavior Architecture" (3 December 2025). Publishes problem-solving commentary on LinkedIn as "Reuben ASK: Insights & Opinions."

Certifications

2020 Internal Auditor: ISO 9001:2015, ISO 14001:2015 and ISO 45001:2018, covering quality, environmental, and occupational health and safety management.
BSI Training Academy, Singapore

2018 Advanced TESOL Certification and Teaching English to Children.
American TESOL Institute, Shanghai

Awards and recognition

2023 Best Student Conference Paper, 2023 NSFC-RGC Conference on Frontiers of Industrial Big Data and Intelligent Systems, Hong Kong.

2020 Best Graduating Student and Valedictorian, Lanzhou University of Technology.

2019 Chinese Government Outstanding International Student Award, Ministry of Education of China: one of 582 international students selected from 127 universities nationwide.

Memberships

Human Factors and Ergonomics Society (HFES)
Member (Emerging Country Membership)

Languages

English (native) · Ewe · Twi · Ghanaian Pidgin English · Mandarin Chinese (professional, HSK 5) · French (elementary)

References available on request.